Modelling In Mechanics

Forces

Weight acts towards Earth. A normal reaction acts perpendicular to a surface. Friction opposes sliding or its tendency. Tension acts along a taut string.

In equilibrium, resultant force is zero. For a particle of constant mass, Newton's second law connects force and motion:

$\mathbf{F} = m\mathbf{a}$.

Motion

Gradient of a displacement–time graph gives velocity. Gradient of a velocity–time graph gives acceleration. Signed area under velocity–time gives displacement; area under speed–time gives distance.

For constant acceleration:

$v = u + at, \quad s = ut + \frac{1}{2}at^2$

$v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u + v)t.$

For variable acceleration:

$v = \dfrac{ds}{dt}, \qquad a = \dfrac{dv}{dt}.$

Common modelling assumptions

Particle: dimensions are negligible; mass is concentrated at a point. Rotation is ignored. Air resistance, when neglected, is a separate assumption.

Smooth surface: no friction. A rough surface can exert friction.

Light, smooth pulley: negligible mass and no friction; an ideal taut light string has equal tension on both sides.

Inextensible string: does not stretch. In a simple fixed-pulley system, connected particles have equal acceleration magnitudes while the string remains taut.

Rod: a rigid body whose transverse dimensions are negligible compared with its length.

Peg or support: fixed; dimensions may be neglected. Whether it is smooth or rough depends on the model.

Always state the assumption that justifies a simplification.

SI units

Quantity	Unit	Symbol
Mass	kilogram	kg
Displacement	metre	m
Time	second	s
Velocity	metres per second	$\mathrm{m\,s^{-1}}$
Acceleration	metres per second squared	$\mathrm{m\,s^{-2}}$
Force	newton	N

Newton's second law gives

$1\,\mathrm{N} = 1\,\mathrm{kg\,m\,s^{-2}}$.

Convert measurements into compatible units before substituting into an equation. Mass is measured in kilograms; weight is a force measured in newtons.

Scalars and vectors

A scalar has magnitude only. Distance and speed are non-negative; mass and time are scalar quantities.

A vector has magnitude and direction. Displacement, velocity, acceleration and force are vectors.

A displacement of $3\mathbf{i} + 4\mathbf{j}$ m has magnitude 5 m.

A particle moves 3 m opposite to the chosen positive direction. State its distance travelled and its displacement.

Distance and displacement

1. A displacement has magnitude 5 m and is directed $60°$ above the positive horizontal axis. Find its components.

1

2. Velocity is $(5\mathbf{i} - 12\mathbf{j})\,\mathrm{m\,s^{-1}}$. Find the speed.

2

3. A force of 8 N acts $45°$ below the positive horizontal axis. Find its vector form.

3

Further Examples 7

1. A force of 6 N acts downwards and to the left, at 60° to the downward vertical. Find its components.

1

2. Find the magnitude of acceleration $(6\mathbf{i} - 8\mathbf{j})\,\mathrm{m\,s^{-2}}$.

2

3. A displacement of 4 m is directed 30° above the negative horizontal axis. Write it using unit vectors.

3

Test Your Understanding 8

1. A man walks from A to B, then to C. His displacements are $\overrightarrow{AB} = (6\mathbf{i} + 4\mathbf{j})$ m and $\overrightarrow{BC} = (5\mathbf{i} - 12\mathbf{j})$ m.

   a. Find the magnitude of $\overrightarrow{AC}$.

   b. Find the total distance walked.

1a

1b

2. A car's velocity is $(3\mathbf{i} - \mathbf{j})\,\mathrm{m\,s^{-1}}$. Find the angle its direction makes with $\mathbf{i}$.

2

2. A car has velocity $(12\mathbf{i} - 10\mathbf{j})\,\mathrm{m\,s^{-1}}$. Find its speed and the angle with $\mathbf{i}$.

2

3. A motorbike has acceleration $(3\mathbf{i} - 4\mathbf{j})\,\mathrm{m\,s^{-2}}$. Find its magnitude and the angle with $\mathbf{j}$.

3

7. A boat has acceleration $(-0.05\mathbf{i} + 0.15\mathbf{j})\,\mathrm{m\,s^{-2}}$. Find its magnitude and the angle with $\mathbf{i}$.

7